Closure as Asymptote: A Category Distinction Between General and Cosmological Theories

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Abstract

The criterion of *closure* — the demand that a physical theory derive all its parameters from a smaller set of first principles — is conventionally applied to general theories of nature (electromagnetism, the Standard Model, general relativity in its single-spacetime applications) and to cosmological theories (ΛCDM and its alternatives) by the same standard. This paper argues that the uniform application embodies a category distinction that is often left implicit. General theories describe a class of systems indexed by parameters; the standard closure operations — inference across instances, and embedding into a larger theory — exploit this class structure. A cosmological theory describes a single object, the universe, and by the encompassing definition of "universe" neither operation is available in principle. The two activities differ not in difficulty but in the inferential resources they have access to. We illustrate the structural point with two parameters whose closure status is unsettled — the cosmological constant Λ in ΛCDM, and a modulation lock $\theta_c = \ln \varphi$ recently proposed in the Lagrangian Variable Cosmology (LVC) working-paper series — and find that both share the same categorical feature. Under an explicitly hypothetical relaxation of the encompassing definition (an "open-universe" thought experiment), parameters that remain primitive in the standard cosmological setting admit derivations at the level of rigour customary in theoretical physics; we work one such derivation in detail. We propose that the appropriate evaluative standard for cosmological theories is partial closure approached as an asymptote — explicit in degree, transparent in primitives, and accountable in direction — rather than closure as a fixed target. The argument does not adjudicate among cosmological proposals, does not propose that the universe actually has an outside, and does not relax the standards of empirical rigour. It clarifies, by structural diagnosis, what the closure standard categorically can and cannot demand of a cosmological theory.

1 Introduction

A physical theory is commonly evaluated, in part, by how many of its parameters are *derived* from how many *posited*. Quantum electrodynamics is regarded as a successful theory in part because two parameters — the electron's charge and mass — together with the field equations generate the entire spectrum of atomic transitions, scattering cross-sections, and radiative corrections that have been measured. The Standard Model of particle physics is regarded as *incomplete* by the same standard: nineteen empirical parameters are required to specify its predictions, and the persistence of this count constitutes the model's central open problem [1].

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The same standard is conventionally applied to cosmological theories. The ΛCDM model fits the present geometry, expansion history, and large-scale structure of the universe with six parameters in its standard parametrisation: H_0, Ω_m, $\Omega_b h^2$, τ, A_s, n_s [2]. Among these, the cosmological constant Λ (entering as $\Omega_\Lambda = 1 - \Omega_m - \Omega_r$) attracts particular attention because its measured value $\Lambda \approx 1.1 \times 10^{-52}$ m^{-2} is not derived from any accepted underlying theory; quantum-field-theoretic estimates of vacuum energy density disagree with the observed value by 60–120 orders of magnitude depending on the regularisation cutoff [3]. The discrepancy, known as the cosmological constant problem, is conventionally read as evidence that ΛCDM, like the Standard Model, has a closure gap that a deeper theory should eventually fill.

Recent alternative cosmologies exhibit the same pattern. In one such proposal, the Lagrangian Variable Cosmology (LVC) series [4, 5, 6], the late-time expansion rate $H(z)$ is modified by a localised multiplicative modulation $T(z) = H(z)/H_{\Lambda\mathrm{CDM}}(z)$ parametrised by a centre coordinate, a width, and an amplitude. In its current full-lock construction, the centre takes the algebraic value $\theta_c = \ln\varphi \approx 0.4812$ in the logarithmic-time coordinate $\theta = \ln(1+z)$, with $\varphi = (1+\sqrt{5})/2$ the golden ratio; the same construction relates the modulation amplitude and width to θ_c by structural relations within the model [4]. The value $\theta_c = \ln\varphi$ is consistent with the free-fit best estimates to four decimal places on the present datasets, and the construction's static-sector Lagrangian admits the observed modulation profile as a classical solution. The *position* θ_c, however, is not derived from the action; it is an empirical input, in the same operational sense in which Λ is an empirical input in ΛCDM.

This paper is not a contribution to the cosmological constant problem. It is not an advocacy for LVC or for any other alternative to ΛCDM. Its question is methodological: *under what conditions does the demand for full parameter closure apply to a cosmological theory, and what evaluative standard takes its place where it does not?*

We argue that the demand, as it is usually formulated, applies less cleanly to cosmological theories than to general theories of nature. The argument rests on a structural difference between the two categories: a general theory describes many instances of the same kind of system, while a cosmological theory describes a single instance — the universe — and no second instance is available either observationally or, by the standard definition of "universe," in principle. The closure operation, which in general theories means *reduction across instances* or *embedding into a larger theory*, has no analogue in cosmological theories under the standard definition.

We further isolate, via a thought experiment, which features of the closure problem are *categorical* (depending on the encompassing definition of "universe") and which are *technical* (depending on the specific structures of the universe under study). When the encompassing definition is hypothetically suspended — when one allows reference to an external structure outside U — parameters that resist closure in the standard setting are found to admit closure at the level of rigour customary in theoretical physics, within at least one natural construction. The contrast is asymmetric and informative. It indicates that closure of cosmological parameters is not blocked, on the available analysis, by technical inadequacy of cosmological theorising but is consistent with a categorical fact: the inferential resources used for closure in general theories are unavailable in principle to a theory of the encompassing whole.

We propose that the appropriate evaluative standard for cosmological theories is not closure as a fixed target but closure as an *asymptote* — partial closure with its degree, primitives, and direction made explicit. This standard preserves the rigour of cosmological theorising while declining to penalise it for failing to reach a limit that, by category, no cosmological theory can reach.

We emphasise at the outset what this paper is not. It is not a critique of ΛCDM; the categorical feature we describe attaches to *every* cosmological theory, including LVC, equally. It is not an advocacy for LVC; the empirical question of whether LVC's modulation describes our universe is independent of the methodological question raised here and is addressed by the usual instruments of cosmological inference. It is not a claim that the universe has, or does not have, an outside; the thought experiment of Section 5 is performed under an explicitly suspended assumption. And it is not a recommendation to relax empirical standards; the asymptotic standard of Section 8 is rigorous, differing from the closure standard in its target rather than its discipline.

The remainder of the paper is organised as follows. Section 2 makes precise the distinction between general and cosmological theories. Section 3 states the closure argument in its conditional form. Section 4 addresses the principal counterexamples (the simulation and multiverse hypotheses). Section 5 conducts the thought experiment under the open-universe assumption and records a worked derivation. Section 6 compares the structural limit to known self-reference limits in formal systems. Section 7 discusses what the argument does and does not imply, including boundary cases. Section 8 develops the asymptotic standard. Section 9 records the limitations of the argument. Section 10 concludes.

2 The Distinction between General and Cosmological Theories

We say a theory T is a **general theory** if T describes a class C of physical systems indexed by parameters and initial/boundary conditions, such that:

(G1) C contains more than one element accessible to observation in principle;

(G2) Predictions of T concerning one element of C are testable, in principle, against observations of other elements of C;

(G3) Reduction of T's parameter count is operationally equivalent to a statement that holds *across* C.

By contrast, T is a **cosmological theory** if T describes a single object U (the universe) such that:

(C1) No second element of the same kind as U is, by the standard meaning of "universe," accessible in principle;

(C2) All observations available for testing T are observations of U;

(C3) Reduction of T's parameter count, if it occurs, must be expressed *within* T's description of U, not across instances.

The distinction is structural. It does not concern difficulty: quantum field theories are computationally far harder than the Friedmann equations. It does not concern precision: cosmological parameters are now measured to fractional precision rivalling many laboratory observables. It concerns what the theory's predictions *answer to*. A general theory answers to a population of systems; a cosmological theory answers to one system.

The Standard Model is a general theory. Its predictions of particle properties are tested in many experiments, across many independent runs of the same experiment, each constituting a separate instance of the kind of system the theory describes. Mendelian genetics is a general theory of organisms; it is tested on many organisms. General relativity, when applied to specific spacetimes — Schwarzschild, Kerr, gravitational waves — functions as a general theory: each spacetime is a distinct instance. When general relativity is applied to *the* universe, with the additional FLRW symmetry and a specific energy content, it becomes part of a cosmological theory: there is now only one instance.

The single-instance character is not a contingent feature that further observation might remedy. It is built into the *definition* of "universe" in the relevant cosmological sense: the universe is the totality of physical entities causally accessible (under one's preferred framework), and in the strongest formulations the totality of physical entities full stop. Two different "universes" in this sense would not be different instances of the same physical kind: either they would be parts of a larger encompassing system (in which case that larger system is the universe), or they would stand in no physical relation to one another (in which case the cosmological theory of one has no observational bearing on the other). The single-instance feature is thus a *category property* of cosmological theories, not an empirical accident open to revision.

Two clarifications are in order. *First*, the FLRW symmetries are approximate: at small scales the universe is highly inhomogeneous, and at the largest scales the homogeneity is itself an empirical claim under continued test [7]. The single-instance feature does not depend on these symmetries being exact; it depends on the universe being one thing, which it remains under any continuous deformation of the symmetry assumption. *Second*, multiverse proposals do not by themselves overturn the single-instance feature; they typically replace "universe" by "multiverse" as the encompassing single object, with the same category property attaching to the new term. We return to this in Section 4.

The distinction is implicitly understood. Cosmologists routinely note that their data set has, in the relevant inferential sense, $N = 1$ [8]. The argument of this paper is that this implicit recognition has direct consequences for what one is entitled to demand of a cosmological theory in the way of closure, and that the consequences are not always made explicit.

3 The Closure Argument

We define **closure** of a theory T with respect to a parameter p as: there exists a derivation, within T's accepted framework, that uniquely fixes the value of p from a strictly smaller set of more fundamental quantities also within T's framework. **Full closure** of T is closure with respect to every parameter of T except a minimum set declared as primitive. **Partial closure** is closure with respect to a proper subset.

The closure operation in general theories proceeds, in standard practice, by either of two routes.

(R1) Inference across instances. A parameter p is shown to take the same value across all instances of the system T describes — itself a non-trivial empirical claim — and a higher-level theory T' is constructed in which p is derived from invariants of T' that account for the across-instance uniformity. The mass of the electron, taken to be the same across all electrons (and verified across a vast number of independent measurements), would be derived in a more fundamental theory from the Higgs mechanism and a Yukawa coupling; the Yukawa coupling itself would then be the deeper input.

(R2) Embedding in a larger theory. A parameter p, undetermined within T, is shown to take a specific value within an embedding theory T'' in which T is a limiting or sectoral case. The Yukawa couplings of the Standard Model are not themselves derived within the SM; a putative grand unified theory or string compactification would embed the SM and derive them from group-theoretic or geometric inputs of T''.

Both routes presuppose, respectively, *across-instance information* or *embedding information.* In a cosmological theory T:

- By (C1), across-instance information is unavailable, by the encompassing definition of "universe."
- By (C2)–(C3), embedding information is unavailable, since the universe is, by the encompassing definition, that which is not embedded in anything physical.

Therefore, in a cosmological theory under standard definitions, the standard closure operations have no resource on which to draw.

This does not establish that closure is impossible *tout court*. A cosmological theory may still derive one parameter from others via a structural relation *internal* to its description of U; such a relation does not require R1 or R2. The Friedmann equation derives the present expansion rate from the present matter and dark-energy densities, given a metric ansatz; this is internal partial closure, and ΛCDM achieves a non-trivial amount of it. But closure of a parameter from *outside* its own theoretical context — the operation that R1 and R2 perform — has no analogue.

We state this formally as the **Closure Argument**:

> **Premise 1.** Closure of a parameter p in a theory T proceeds, in standard practice, by either inference across instances (R1) or embedding in a larger theory (R2).
>
> **Premise 2.** A cosmological theory T describes a single instance U; no second instance is available in principle, by the encompassing definition of "universe."
>
> **Premise 3.** A cosmological theory T describes the totality of physical entities (in its encompassing sense); no larger theory in which T is a sub-theory exists in principle, by the same definition.
>
> **Conclusion.** A cosmological theory T cannot achieve closure by R1 or R2. Any closure it achieves must proceed by structural relations internal to its description of U.

The conclusion is conditional on the encompassing definition. Section 4 addresses the principal candidate counterexamples to this definition.

4 The Simulation and Multiverse Counterexamples

The most discussed counterexample to the encompassing definition is the **simulation hypothesis** [9]: that what we observe and inhabit is a simulation running on a substrate external to it. If true, "our universe" would have an outside (the substrate), and the encompassing definition would fail for it.

We note: if a system S is simulated on a substrate X, then X is either itself a universe in the encompassing sense (a totality of physical entities) or part of a larger such system. The cosmological theory of *that* larger system inherits the single-instance feature — by definition, it has no outside. The simulation hypothesis therefore does not eliminate the cosmological category; it *relocates* it. The cosmological theory of the simulated S becomes a *partial* theory (describing a sub-system, akin to the theory of a planetary atmosphere); the cosmological theory in the categorically distinct sense is the theory of the encompassing system X.

This observation generalises. Any proposal that places "our universe" within a larger encompassing structure thereby identifies a new candidate for the term "universe" in the cosmological sense. The category property of single-instance-ness attaches to whatever is the largest encompassing system in the proposal. To dissolve the category one would have to deny that there is *any* largest encompassing system — that is, deny that the totality of physical entities exists as a coherent referent. Whether such a denial is intelligible is a question we set aside; we observe only that the cosmological category, as defined in Section 2, is robust against the standard moves that appear to threaten it.

Multiverse proposals are addressed by the same observation. A multiverse containing many "universes" is itself an encompassing single object in the relevant sense; the cosmological theory of the multiverse inherits the single-instance feature. A cosmological constant of our local sub-universe might be derived, within a multiverse theory, from selection effects [10]; but the multiverse itself, *as the encompassing structure*, then carries parameters that are not so derivable. The category has been pushed one level up, not dissolved.

This does not impugn multiverse theorising. It clarifies what multiverse theorising achieves: it converts a closure problem at the level of our local sub-universe into a closure problem at the level of the multiverse, via an R2-style embedding. The conversion is real progress *within the partial-closure standard* developed below (Section 8); it is not full closure, because the encompassing structure remains.

5 The Open-Universe Thought Experiment

To clarify the categorical character of the closure limit, we conduct a thought experiment in which the encompassing definition is *temporarily suspended for the sake of inquiry.* We are not proposing that the universe has an outside. We are isolating which features of the closure problem are *categorical* — depending on the encompassing definition — versus *technical* — depending on the specific structures of the universe under study.

5.1 Setup

Suppose, as a working assumption only, that the universe U is embedded in a larger structure B with which U is in some specified physical relation. We deliberately leave the interpretation of B underdetermined. B may be construed as

- a higher-dimensional bulk in which U is a brane (as in Randall–Sundrum constructions [11]);
- a boundary condition imposed on the asymptotic structure of U;
- a meta-structure containing U as a sub-region (a "multiverse" with explicit dynamics);
- a gravitational matrix in which U is locally distinguishable.

The interpretation matters for the specific physical content but not for the structural point. The thought experiment requires only that B carry physical content from which inferences into U are possible. We refer to this as the **open-universe assumption**, in contrast to the standard **closed-universe assumption** under which U has no physical outside.

5.2 Worked example: closure of $\theta_c = \ln \varphi$ under the open-universe assumption

We work one example in detail to confirm that, under the open-universe assumption, closure of a cosmological parameter that is primitive in the closed-universe setting becomes technically feasible. We use the modulation centre $\theta_c = \ln \varphi$ of the LVC construction [4] because it has a sharp algebraic structure that lends itself to a clean derivation; nothing in the structural point depends on this specific example.

Suppose B is a five-dimensional hyperbolic bulk H^5 with constant sectional curvature $K = -1$ in natural units, and U is a four-dimensional brane embedded in B at a $\mathbb{Z}_2$ fixed locus of a reflective symmetry of B. The geometry of B and the embedding then have two structural consequences.

Fact 1: Killing spinor eigenvalue on H^n. A Riemannian or Lorentzian space of constant negative curvature K admits Killing spinors ψ satisfying

$$\nabla_\mu \psi = \lambda\, \gamma_\mu\, \psi. \tag{1}$$

The integrability of this equation on a maximally symmetric space gives, from $R^{\mu\nu}{}_{\rho\sigma} = K(\delta^\mu_\rho \delta^\nu_\sigma - \delta^\mu_\sigma \delta^\nu_\rho)$ and the identity $[\gamma^\rho, \gamma^\sigma] R_{\mu\nu\rho\sigma} \propto R_{\mu\nu}$, the condition $\lambda^2 = -K/4$. With the curvature scale chosen as $K = -1$, the Killing eigenvalue magnitude is therefore $|\lambda| = \frac{1}{2}$. The choice of curvature scale is conventional, but the *ratio* between the Killing eigenvalue and the curvature scale is geometric, fixed by the integrability of the Killing-spinor equation [12].

Fact 2: BPS-saturated brane at a $\mathbb{Z}_2$ fixed locus. A codimension-one brane at a $\mathbb{Z}_2$ fixed locus of a bulk reflective symmetry that preserves half the bulk Killing spinors is the standard half-BPS brane configuration [13]. The surviving spinor at the brane satisfies $\gamma_\perp \psi|_{\text{brane}} = +\psi|_{\text{brane}}$, where $\gamma_\perp$ is the gamma matrix associated with the brane's normal direction. Parallel transport of the Killing spinor along a geodesic from a reference point at hyperbolic distance θ_c then implies, with the curvature normalisation of Fact 1, that the spinor's hyperbolic momentum at the brane saturates the BPS bound at

$$p_{\text{brane}} = \tfrac{1}{2}\, m_{\text{bulk}}, \tag{2}$$

where m_{bulk} is the bulk mass scale (unique in the gravitational sea interpretation, where B and U share the same gravitational coupling and there is no separate brane Planck scale). In natural units where $m_{\text{bulk}} = 1$, this reads $p_{\text{brane}} = \frac{1}{2}$.

Combining. Hyperbolic-spinor kinematics on B (with curvature normalised to unity) gives $p = \sinh(\theta)$, where θ is the hyperbolic geodesic distance from the reference point. The brane position θ_c is therefore determined by

$$\sinh(\theta_c) = \tfrac{1}{2}. \tag{3}$$

The positive solution is

$$\theta_c = \operatorname{arcsinh}\!\left(\tfrac{1}{2}\right) = \ln\!\left(\frac{1+\sqrt{5}}{2}\right) = \ln \varphi. \tag{4}$$

Position of alternatives within the setup. Other values of θ_c that one might consider — corresponding to alternative settings of the BPS condition — sit differently relative to the structural premises chosen above. The value $\theta_c = \ln(1+\sqrt{2})$ (the silver mean, corresponding to $\sinh(\theta_c) = 1$) would require doubled BPS saturation rather than the half-BPS configuration of Fact 2, and is therefore not the configuration most naturally produced by a $\mathbb{Z}_2$ fixed-locus brane preserving half the bulk Killing spinors. Values of $\sinh(\theta_c)$ of the form $3/2$, $5/2$ and so on would correspond to non-minimal fermionic representations (spin $3/2$ or higher), whereas the Killing spinor on H^n is in the minimal spin-$\frac{1}{2}$ representation; this is a separability rather than an exclusion, but it is a structural distinction. Transcendental values such as π and e are not of the form arcsinh of a rational; since the construction above produces a hyperbolic angle equal to arcsinh of a rational (the spinor charge in natural units), such values do not arise in this construction. Within one natural class of open-universe constructions — those built from a hyperbolic bulk, minimal Killing-spinor representation, and half-BPS embedded brane — $\ln\varphi$ thus emerges as a distinguished value consistent with the structural premises. We do not claim that this is the unique class of open-universe constructions that could be considered, nor that the structural premises above are the only natural premises one might adopt; we claim only that within this class, $\ln\varphi$ is the natural outcome.

5.3 The level of the derivation

The derivation is at the level of rigour standard in theoretical physics. Each step rests on a well-defined geometric or algebraic fact (Killing-spinor integrability, $\mathbb{Z}_2$ half-BPS construction, hyperbolic-spinor kinematics). The construction relies on a specific class of open-universe setups; within that class, the structural premises pick out $\ln\varphi$ rather than a fitted value. The construction does not claim that this class exhausts the natural open-universe setups, nor that the residual choices within the class (the specific interpretation of B, the dimensionality of the bulk, the precise form of the BPS algebra) are themselves derived.

The derivation is *not* at the standard of pure mathematics: it does not single out the open-universe setup uniquely among all logically possible setups, and the choice of setup family is itself a modelling choice rather than a forcing. This is the standard situation in theoretical physics; analogous derivations of mass relations in extra-dimensional models, central charges in conformal field theory, or critical exponents in renormalisation-group analyses share this character. The structural point of this paper — that closure becomes technically feasible under an open-universe assumption when it is not feasible under a closed-universe assumption — does not depend on the specific construction being the uniquely correct one. It depends only on *some* such construction existing.

5.4 The contrast with the closed-universe setting

We emphasise the structural feature of the derivation: it relies *entirely* on the existence of B. The Killing spinor eigenvalue $\frac{1}{2}$ is a property of the bulk's intrinsic curvature; the BPS saturation condition is a property of how U sits inside B; the spinor momentum at the brane is a property of how the bulk spinor restricts to U. Remove B, and every premise of the derivation is lost.

In the closed-universe formulation, the parameter θ_c remains a primitive input. It can be fitted, and structural relations among other parameters can be imposed and tested, but no derivation route in the senses (R1) or (R2) of Section 3 is available. The asymmetric outcome is not, on the available evidence, a contingent failure of inquiry — attempts to derive $\theta_c = \ln\varphi$ from closed-universe structures spanning a number of candidate frameworks (Fibonacci/$\mathrm{SU}(2)_k$ embeddings, KAM stability, loop-quantum-gravity Immirzi parameter, dS/CFT primary weights, quasicrystalline lattices, and others) have to date not produced a forcing of the value within any of those frameworks [5, 6]. Each candidate either produced a near-neighbour value (silver mean, $2\ln\varphi$) or required ad hoc inputs equivalent to imposing the answer. We do not claim that no closed-universe forcing of θ_c could ever be found; we observe that the categorical structure of Section 3 explains the systematic difficulty: the inferential resource that would close θ_c (an embedding bulk and its Killing-spinor structure) is by definition unavailable in a closed-universe theory.

5.5 The same structure applies to Λ

The same categorical pattern holds for the cosmological constant Λ. Closed-universe attempts to derive Λ from quantum-field-theoretic vacuum-energy calculations disagree with the measured value by 60–120 orders of magnitude [3]. Open-universe approaches — multiverse anthropic selection [10], the string landscape [14], holographic boundary conditions [15] — typically achieve closure conditional on an embedding structure (a multiverse, a flux compactification, a holographic dual). The conditional character of these proposals is consistent with the structural diagnosis of Section 3: closure proceeds when an embedding becomes available, and not otherwise.

The structural symmetry between the Λ case and the θ_c case is the central observation. The two parameters belong to different cosmological models, are of different magnitude scales, and address different aspects of the late-time expansion history. They share, however, exactly the closure-status feature that the categorical argument predicts: each is a primitive input in its closed-universe model, and each becomes derivable under an open-universe embedding. Neither model is more or less afflicted by the closure problem than the other; both exhibit the same categorical feature, which is a property of cosmological theorising as such, not of either model.

5.6 What the thought experiment shows

The thought experiment is, by construction, not a physical hypothesis. We have not claimed that the universe has an outside; the open-universe assumption was suspended for the sake of inquiry. What the thought experiment does show is:

- A parameter that, on the available evidence, resists derivation in a closed-universe theory admits derivation when an outside is supplied — at the level of rigour customary in theoretical physics, within at least one natural construction.
- The closure obstacle in the closed-universe case is therefore not, on the structural analysis, a *technical* deficit of cosmological theorising; it is consistent with a *categorical* consequence of working without an outside.
- The standard demand for closure, formulated for general theories (which always have R1 or R2 available), is asking a cosmological theory for an inferential resource that the cosmological category does not provide.

We refer to the third point as the **category trap**: the application to a cosmological theory of an evaluative standard whose underlying preconditions (the availability of R1 or R2) do not categorically obtain.

A secondary observation, which we record but do not develop further here, is that the open-universe derivation in Section 5.2 indicates the LVC modulation centre is not *intrinsically* underivable: it admits at least one natural derivation when an embedding is supplied, while remaining a primitive input in the closed-universe formulation in which LVC, like ΛCDM, is operationally set. Whether and how the LVC programme might in future construct a closed-universe derivation of θ_c through internal structural relations alone is a question for the LVC literature; the present paper makes no claim on the outcome and is fully neutral on the empirical status of the LVC modulation.

6 Comparison with Known Self-Reference Limits

The structural feature we have identified — that a theory describing a totality is barred from a derivation route available to theories describing parts — is analogous in form to limits established for formal systems by Gödel [16] and Tarski [17]: a formal system rich enough to encode arithmetic cannot prove its own consistency from within, and a sufficiently expressive formal language cannot define its own truth predicate.

We do not assert that Gödel's or Tarski's theorems apply *as theorems* to physical theories; they apply to formal systems of a specific kind, and their adaptation to physical cosmology would require careful formalisation that we do not undertake here. We assert only that the *categorical structure* is similar: in both cases, an operation that is available across systems of a class is unavailable to the maximal system that includes all of them. Closure across instances (in physical theories) and proof from outside the system (in formal systems) share this feature. We use the analogy heuristically to indicate that the limit we describe is a recurring structural pattern, not an idiosyncrasy of cosmology.

A more direct physical analogue is the impossibility of measuring the absolute position or velocity of an isolated system. Features that would distinguish *one position from another* require an external reference; an isolated system has none, and absolute position is therefore not a physical observable. The relation of a cosmological theory to closure is structurally parallel: features that would distinguish "the universe with this Λ" from "the universe with a different Λ" would require something to be different elsewhere; for the encompassing universe, there is no "elsewhere." This is not a poetic statement; it is a description of what the encompassing definition entails.

7 Implications and Limitations of Application

We state in plain form what the categorical-distinction thesis does and does not imply.

7.1 What it does not imply

- It does **not** imply that closure is impossible *tout court* for cosmological parameters. Structural relations internal to U may yet derive some parameters from others. The argument concerns only the routes (R1) and (R2).
- It does **not** imply that the search for deeper theory in cosmology is misconceived. Pursuing greater partial closure is fully consistent with — indeed, recommended by — the asymptotic standard of Section 8.
- It does **not** imply that ΛCDM, LVC, or any other cosmological model is flawed for failing to derive its parameters from first principles. The category property entails only that the standard demand for full closure is not entitled to its usual evaluative force in this category.
- It does **not** imply that the universe has, or does not have, an outside. The thought experiment of Section 5 was conducted under an explicitly hypothetical assumption; we draw no conclusion about the actual physical situation.
- It does **not** recommend any relaxation of empirical rigour. The asymptotic standard is rigorous; it differs from the full-closure standard in target, not in discipline.
- It does **not** privilege any one cosmological model over another. Λ in ΛCDM and $\theta_c = \ln \varphi$ in LVC are treated on the same footing throughout, and the structural analysis applies to both equally.

7.2 What it does imply

- The demand for full closure, when applied uniformly to general and cosmological theories, embodies a category distinction that has not always been made explicit. The structural difference between the two cases is consequential.
- The appropriate question for a cosmological theory is not "does it close?" but "to what degree, and along what direction, has it achieved partial closure, and how transparently are its primitive inputs declared?" (Section 8).
- Cosmological parameters that resist closure within their respective models — Λ in ΛCDM, $\theta_c = \ln \varphi$ in LVC, and others — are not anomalies of those models; they exhibit a structural feature of cosmological theorising itself.
- Conversely, a cosmological parameter that *is* derived within its model — by an internal structural relation — exhibits genuine partial closure, and is evaluable accordingly within the asymptotic standard.

7.3 Boundary cases

The categorical distinction is not absolute. Two cases on the boundary merit explicit acknowledgement.

First, if future cosmological work were to firmly establish that the universe is *not* the encompassing structure — that some larger physical system B definitely exists and is observationally accessible in some way — then "the universe" in the cosmological sense relocates to the encompassing system that includes B. The single-instance category property transfers with the relocation; closure of parameters of our local sub-universe via inferences from B becomes technically possible, but the theory of *the whole* (including B) inherits the categorical limit. The distinction we have drawn applies to whichever description plays the encompassing role.

Second, if future cosmological work were to firmly establish that the encompassing structure does not exist — that the totality of physical entities is not a coherent referent — then the cosmological category as we have defined it dissolves. We have not argued against this possibility; we observe only that standard cosmological practice, including the practice of fitting models such as ΛCDM and LVC to data, presupposes that there is *some* maximal physical system to be modelled. If that presupposition fails, the closure question changes character, and the standard for evaluating cosmological theories changes with it; the category trap then does not arise, because the category itself has been abandoned.

In both boundary cases, the present analysis is *conditional* on the standard meaning of "universe" remaining the operative one. Within that meaning, the analysis stands.

8 Closure as Asymptote: A Standard for Cosmological Theories

We propose that cosmological theories be evaluated against the standard of **asymptotic closure**: closure approached as a limit, with the degree of approach made explicit. The proposal has four components.

8.1 Partial closure is the operational goal

A cosmological theory T should be evaluated by the proportion and the structural depth of its closure, not by whether it has reached full closure. Of T's parameters, some will be derived from others by structural relations internal to T (such derivations are admissible without R1 or R2); others will remain primitive inputs. The proportion derived, weighted by the depth of the derivations, is one measure of T's partial closure.

ΛCDM achieves partial closure at the level of, for example, deriving the matter-radiation equality redshift from Ω_m, Ω_r, and the recombination physics, and deriving the present expansion rate from the present matter and dark-energy densities via the Friedmann equation. These are structural relations internal to the model. ΛCDM does not derive Λ from the other parameters, nor does it derive Ω_m from anything more fundamental within its accepted framework; both remain primitive.

LVC achieves partial closure at the level of, for example, the structural relation between the modulation width and amplitude ($w = A/\ln\varphi$) and between the modulation centre and the matter–dark-energy equality redshift ($\ln\varphi = \varphi \cdot \theta_{\mathrm{eq}}$) within its full-lock construction [4]. These relations are internal to the LVC framework and reduce its parameter count relative to an unrestricted parametric fit. LVC does not derive the absolute value $\theta_c = \ln\varphi$ from its action; the value is a primitive input on the same operational footing as Λ in ΛCDM.

Both models exhibit non-trivial partial closure. The comparative *extent* of their partial closures is a function of how many parameters each model contains primitively and how much of its parameter space is structurally constrained by relations within it. This is an evaluable quantity, not a verdict; both models can in principle improve their partial closure by future work.

8.2 Transparency about irreducible inputs

A cosmological theory ought to declare its irreducible inputs explicitly. ΛCDM has historically been transparent in this respect: Λ, Ω_m, $\Omega_b h^2$, τ, A_s, n_s are listed as primitive in standard parametrisations [2], and proposed derivations are clearly identified as proposals rather than parts of the model. LVC has, in its working-paper series, similarly declared its irreducible inputs: the value of θ_c (currently $\ln\varphi$ on phenomenological grounds), the modulation amplitude scale, and the gravitational coupling are stated as inputs, with the derived relations clearly separated from the imposed ones [4, 5, 6].

Transparency, here, is the precondition for evaluating partial closure: opacity about which inputs are primitive prevents the comparison. A theory that *appears* to have fewer primitives because it has not declared all of them is not more closed; it is less transparent.

8.3 Direction of closure effort

The *direction* in which a cosmological theory is moving — which parameters it is attempting to close next, by what proposed structural relations — is itself an evaluable feature. A theory whose closure direction is *internally* structured (proposing relations among its existing parameters within a single framework) is in a different epistemic position from one whose closure direction is *externally* directed (relying on hypothetical embedding structures). Both are legitimate; neither is exclusively correct. The asymptotic standard does not prefer one direction; it requires only that the direction be made accountable.

8.4 The asymptote is not a target

The four components above are operational. They permit the comparison of cosmological theories on closure-related grounds without requiring any of them to achieve a limit that no cosmological theory can categorically reach. The full closure of a cosmological theory is not a *goal* in the sense in which closure of a general theory may be a goal; it is an *asymptote* — a line approached without being touched. The work of moving toward the asymptote is meaningful and evaluable; the demand that the asymptote be reached is a category error.

We do not propose the asymptotic standard as a replacement of empirical adequacy. A cosmological theory must still fit data, predict new observables, and survive falsification on its empirical claims. The asymptotic standard governs only the *closure dimension* of evaluation, which has historically been one (often implicit) dimension among several.

9 Limitations of the Present Argument

We list the limitations of the argument as presented.

- **Limit 1: Definition-dependence.** The Closure Argument rests on the encompassing definition of "universe" (Section 2). If a different definition is adopted — for example, restricting "universe" to the observable patch with explicit allowance for an external structure — the conclusion may not follow. We have argued (Sections 4, 7) that the standard definition is robust against the usual challenges, but the dependence is real.
- **Limit 2: Quantitative measure of partial closure.** The asymptotic standard (Section 8) is qualitative in the present formulation. A formal measure of partial closure — e.g., an information-theoretic compression ratio, a derivation-tree depth, or a Kolmogorov-style complexity — would be needed to make the standard fully operational for Bayesian model comparison. We do not provide such a measure here.
- **Limit 3: Single worked example.** The thought experiment of Section 5 uses one specific example ($\theta_c = \ln \varphi$). A second worked example, for instance an open-universe derivation of Λ from a specific embedding, would strengthen the structural point. Several such derivations have been proposed in the literature [10, 14, 15]; we have not surveyed them in detail.
- **Limit 4: Analogy with formal self-reference.** The comparison to Gödel and Tarski in Section 6 is heuristic; we have not derived a formal theorem about cosmological theory analogous to the incompleteness theorems. Whether such a theorem can be derived under suitable formalisation is open.
- **Limit 5: Emergence-of-spacetime frameworks.** The argument addresses multiverse and simulation hypotheses but does not engage in detail with frameworks in which spacetime itself is emergent (causal set theory, AdS/CFT-inspired emergence, and others), in which the *very status* of the encompassing definition may be at stake. These frameworks may, on a sufficiently developed account, dissolve the category we have defined; we have not surveyed them.
- **Limit 6: Practical implications for working cosmologists.** The argument is about the *evaluation* of cosmological theories; it does not by itself recommend specific changes to data-analysis practice, prior choices, or model-selection criteria. Practical implications of the asymptotic standard, especially for Bayesian model comparison and information-criterion-based selection, are not developed here.

These limits do not undermine the structural distinction of Section 2 or the conditional argument of Section 3; they mark the points at which the present paper sets up questions for further work rather than answers them.

10 Conclusion

The criterion of full parameter closure, applied uniformly to general theories of nature and to cosmological theories, embodies a category distinction that has not always been made explicit. General theories have, by construction, access to inferential resources — inference across instances, and embedding in a larger theory — that cosmological theories, by the encompassing definition of "universe," do not have. The application of the full-closure demand to a cosmological theory therefore asks for a resource the cosmological category does not possess.

The categorical character is illustrated by a thought experiment in which the encompassing definition is temporarily suspended. Under this suspension, parameters that resist closure in the standard cosmological setting admit derivation at the level of rigour standard in physics. We worked one such derivation, for the modulation centre $\theta_c = \ln \varphi$ of the LVC construction, showing that within one natural class of open-universe constructions — a hyperbolic bulk with a $\mathbb{Z}_2$ half-BPS brane carrying a minimal Killing spinor — $\ln \varphi$ emerges as a distinguished value consistent with the structural premises. The same structural pattern accounts for the conditional success of open-universe approaches to the cosmological constant Λ: anthropic selection, the string landscape, holographic embedding. In each case the *type of resource* required for closure is precisely the type the cosmological category, in its encompassing definition, excludes.

The two examples — Λ in ΛCDM and $\theta_c = \ln \varphi$ in LVC — are not in competition in this paper, and the paper makes no claim about which (if either) describes our universe. They illustrate the same categorical fact, which attaches to the *kind* of theory cosmology is, and not to any particular cosmological model.

The appropriate evaluative standard for cosmological theories is partial closure, made explicit in degree, transparent in primitives, and accountable in direction; approached as an asymptote rather than chased as a target. Within this standard, the search for deeper theory and the demand for greater closure remain legitimate and rigorous activities. What is set aside is only the implicit expectation that the asymptote itself can be reached — an expectation that, by the structural argument above, no cosmological theory can satisfy.

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